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1. A Hospital has patient records for many years. How can Data Science be applied?

Ans Data Science can be used to analyse patient records collected over many years. it can identify patterns and trends in diseases, treatments, and patient outcomes. This information can help hospitals understand patient needs and improve healthcare services. it also supports better decision-making based on useful information from the data.

2. A bank wants to detect suspicious transactions. How can Data Science be useful?

Ans Data science can analyse large amounts of bank transaction data to find unusal patterens. it can identify transactions that are different from a customer's normal behaviour. Such analysis can help detect possible fradulent or suspicious transactions. This improves security and helps the bank take appropriate action.

3. A transport department wants to reduce traffic congestion. How can Data Science be applied?

Ans. Data Science can analyse traffic data collected from roads, vehicles, and other sources. it can identify traffic patterns, freauventlly congested areas, and busy periods.

The transport department can use these insights to improve routes and traffic management. This can help reduce congestion and make transportation more efficient.

4. A school wants to improve student performance. mention one benefit of Data science in this case.

Ans. Data Science can analyse students marks and performance data to identify weak areas. it can help teachers understand the performance patterns of individual students. The information can be used to provide suitable support and improve learning outcomes. Thus, Data science helps the school make better decisions using student data.

5. A company has data in Excel tables, customer reviews, images, and videos. which facet of data is shown here?

Ans. The Big Data characteristic shown here is variety. variety refers to the different forms and types of data available for analysis. examples include structured tables, text such as customer reviews, images, and videos. Thus, the presence of different data formats represents variety.

6. A university stores students details in rows and columns. what type of data structure is this?

Ans The data structure is called structured data. structured data follows a fixed format or schema and is usually arranged in rows and columns. examples include data stored in Excel spreadsheets and conventional tables.

Therefore, Student details stored in rows and columns are structured data.

7. A company collects a very large amount of daily user activity data from its mobile app. which Big Data characteristic is applied?

Ans. The Big Data characteristic involved in volume.
Volume refers to the amount of data that is generated, collected, or stored. A mobile application generating a very large amount of user activity data represents high volume. Therefore, this situation demonstrates the volume characteristic of Big data.

8. A business wants to convert customer data into useful sales decisions. which Big Data Characteristic is important?

Ans. The important Big Data characteristic is value.
value refers to the useful insights and benefits that can be obtained from data. A business can analyse Customer data to obtain useful information for making sales decisions. Therefore, value helps converts data into meaningful information for decision making.

9. A company wants to process large datasets using parallel computations. which Big Data tool can be used?

Ans A distributed processing tool such as Hadoop can be used for this purpose.

it supports processing large datasets by dividing large tasks across multiple computers. This allows different parts of the data or tasks to be processed in parallel. Thus, distributed processing helps analyse large datasets more efficiently.

10. A Company wants to visualize Big Data results for decision-makers. which tool or method can be applied?

Ans. The Company can we can use Data visualization to present the results clearly. Charts, graphs, and dashboards can be used to represent patterns and useful information. visualization makes complex data easier to understand for decision-makers. It helps convert analysed data into meaningful information for better decisions.